

Performance Testing

Date	6 July 2026
Team ID	SWTID-2026-7907
Project Name	EduGenie – Google Gemini Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing, Response Time Testing
Target Module / API	Google Gemini API, AI Question Answering, Quiz Generator
Test Environment	Local System (Windows 11, Python Flask)
Test Date	4 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User login Authentication	10	60 seconds	Successful login without errors
2	AI Question Answering	20	120 seconds	Fast and Accurate AI responses
3	Quiz Generation	15	90 seconds	Quiz generated successfully
4	Text Summarization	20	120 seconds	Summary Generated Within expected response time

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 seconds	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Within limit
3	Throughput (Req/sec)	30 Req/sec	35 Req/Sec	Pass	Stable performance
4	Error Rate	< 1%	0.2%	Pass	Very low errors
5	CPU Utilization	< 80%	58%	Pass	Normal usage
6	Memory Utilization	< 80%	63%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings

The EduGenie Application handled Multiple user request efficiently. AI Question answering quiz generation and text summarization responded within Acceptable time limits With stable performance.

Bottlenecks Identified

- Slight delay during simultaneous AI requests.
- Internet speed affected Google Gemini API response time.

Optimized Steps Taken

- Optimized API Calls
- Improved Backend Request handling
- Reduced Unnecessary processing
- Optimised Generation for Faster AI responses

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

